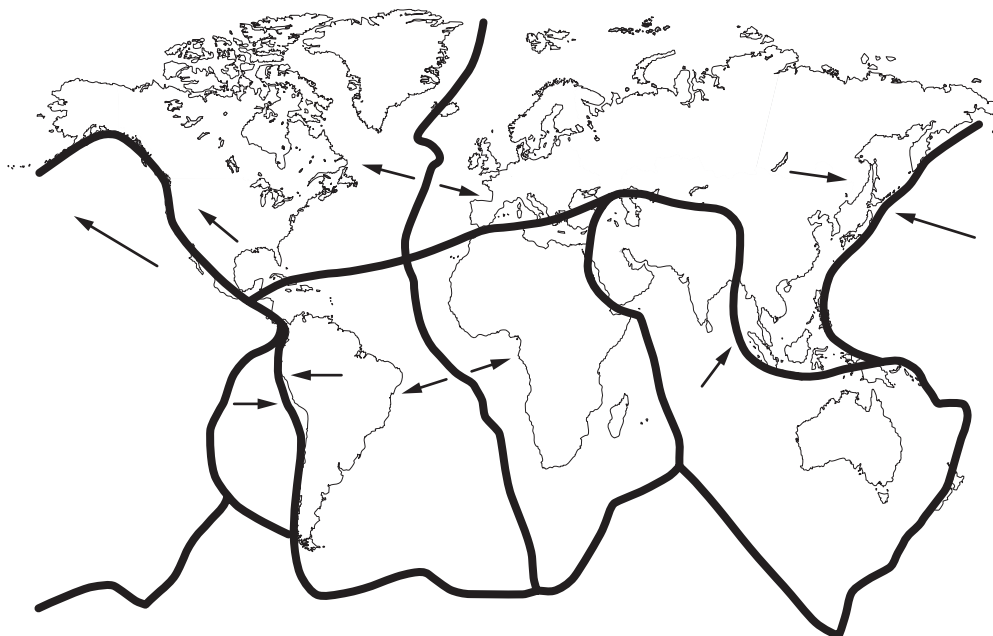


Examiner
only

- 9.** The diagram shows the movement of the major tectonic plates.



Explain what happens at destructive and constructive plate boundaries. Diagrams may be included in your answer.

[6 QER]

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		

THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																									
7 Li Lithium 3		9 Be Beryllium 4																							
23 Na Sodium 11		24 Mg Magnesium 12																							
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30												
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54						
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86						
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																					
Key																									

1
H
Hydrogen
1

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – **NEW**

3410UA0-1



S18-3410UA0-1

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and
Essential Resources

HIGHER TIER

WEDNESDAY, 13 JUNE 2018 – MORNING

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper
 you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen
 or correction fluid.

Write your name, centre number and candidate number
 in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.
 If you run out of space, use the additional page at the back
 of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	6	
2.	9	
3.	5	
4.	9	
5.	11	
6.	6	
7.	10	
8.	10	
9.	8	
10.	6	
Total	80	

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9. The following table shows the decomposition temperatures of Group 2 metal carbonates and nitrates.

Metal	Temperature at which the carbonate decomposes (°C)	Temperature at which the nitrate decomposes (°C)
magnesium	117	89
calcium	178	561
strontium	235	570
barium	267	700

(a) Describe the trends in the stabilities of the Group 2 carbonates and nitrates. [3]

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(b) When a carbonate decomposes it produces carbon dioxide gas. Describe an experiment that could be carried out to show that carbon dioxide gas is produced during the reaction. [2]

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(c) When calcium nitrate decomposes it forms calcium oxide, oxygen and nitrogen dioxide, NO₂.

Write a **symbol** equation for the reaction. [3]

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8



Examiner
only

9. A student is investigating the relative reactivities of the halogens using the elements and solutions of their halides. She adds each halogen to separate solutions of the other halides.

State and explain the observations made. How can the results be used to determine the relative reactivities of the halogens? Include equations in your answer. [6 QER]

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6



Examiner
only

9. (a) Give the observations made during the reactions of lithium, sodium and potassium with water. Explain the trend in reactivity in terms of electronic structure. [6 QER]

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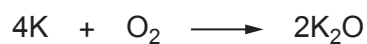
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- (b) Potassium reacts with oxygen to produce potassium oxide.



- (i) Calculate the maximum mass of potassium oxide that could be produced when 15.6 g of potassium reacts completely with oxygen. [3]

$$A_r(\text{K}) = 39 \qquad A_r(\text{O}) = 16$$

Mass = g

- (ii) Another reaction requires 0.050 mol of oxygen gas. Calculate the number of molecules in 0.050 mol of oxygen gas. Give your answer in **standard form**. [2]

$$\text{Avogadro's number} = 6.0 \times 10^{23}$$

Number of molecules =



9. (a) A student tested compounds **A**, **B** and **C** to identify them.

Her observations are recorded in the tables below.

Compound A	Observation	Ion present
Flame test	apple-green flame	barium
Add silver nitrate solution	white precipitate	chloride

Compound B	Observation	Ion present
Flame test	lilac flame	
Add silver nitrate solution	cream precipitate	

Compound C	Observation	Ion present
Flame test	brick-red flame	
Add silver nitrate solution	yellow precipitate	

- (i) Use your knowledge of the tests for ions to complete the tables for compounds **B** and **C**.

[3]

- (ii) Write the chemical formula for compound **A**.

[1]

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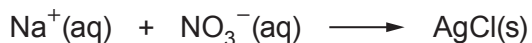
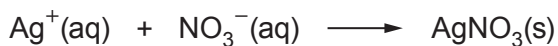
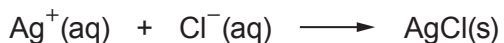
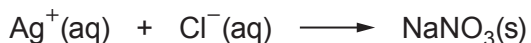


- (b) When silver nitrate solution reacts with sodium chloride solution to produce a white precipitate the following reaction occurs.



Put a tick (✓) in the box which shows the ionic equation for the formation of the precipitate.

[1]

☐☐☐☐☐

Examiner
only

- (c) When silver carbonate is heated it breaks down to give silver metal and carbon dioxide and oxygen gas.



Calculate the mass of silver that would be produced from heating 13.8 g of silver carbonate.

[3]

$$M_r(\text{Ag}_2\text{CO}_3) = 276 \qquad A_r(\text{Ag}) = 108$$

Mass = g

8

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1

2

3

4

5

6

7

0

1

H

Hydrogen

1

Periodic Table of Elements																	
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
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223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89	Key														

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number